Edit, Don't Write: Testing a Writing Skill for Language Models Against Real Human Writing in Five Genres

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Abstract

Advice for making language-model prose read as human usually means banned-word lists or "humanizer" rewriters, which treat the surface as the problem and start from model text. A common request starts from a person's own text: "Can you improve it?" We test a prompt-level writing skill against real writing from before November 2022 in five genres: admitted college essays and short answers, winning science-fair abstracts, Turkish charity runners' appeals and English learners' essays. The skill puts content before vocabulary, forbids invented facts and minimally edits human text. Blind model judges, mostly the generator itself, rated single texts. Written from notes, held-out model texts stayed separable from real ones (AUC 0.90 to 1.00); earlier versions of the skill (7 to 10.1) raised judges' estimates of human authorship only for admitted essays and appeals. Asked to improve held-out real abstracts, appeals and learner essays, plain generation changed 69% to 93% of words and the released skill (version 13.1) 8% to 21%. Paired by text, judges' estimates for the skill's edits were 32 to 48 points (of 100) above plain generation's and 1.4 to 4.1 below an errors-only copyedit's (detectably only on abstracts). For learners, that copyedit itself cost 38 points against the untouched essay. We release the skill and code.

1 Introduction

Experienced readers have learned to recognize machine-written prose. Words such as *delve* rose sharply after 2022 (Kobak et al., 2025; Liang et al., 2024), and five heavy users of these tools, voting together, misclassified only 1 of 300 articles (Russell et al., 2025).

Two responses dominate practice. The first is the banned-word list, distributed in prompts, style guides and community "skills". The second is the commercial humanizer, which rewrites AI-generated text to avoid automated or human detection (Roe et al., 2026). Each treats the surface as the problem. Wikipedia's editors, whose guide the most-starred humanizer skill is built on (Chen, 2026), warn against exactly this: "Please do not merely treat these signs as the problems to be fixed; that could just make detection harder" (Wikipedia contributors, 2026). Both responses also ignore a second failure of model prose: models fill gaps with plausible details nobody supplied (Ji et al., 2023).

These responses concern text that a model wrote. A common situation gets less attention: a person who wrote a text asks a model to improve it. Grammar-only AI help can already shift meaning (Abdulhai et al., 2026), and heavy rewriting erases the signals that identify an author (Malik and Awan, 2026). A student who must submit a science-fair abstract or an application as their own work needs the edited text to stay theirs.

We built a prompt-level writing skill (a set of instructions an agent loads when a task matches) that audits content before vocabulary, is told never to add a fact the writer did not supply, and makes minimum edits to human text. In model-versus-model rankings on invented briefs, its version 4, chosen over version 6 after the held-out round, ranked above plain generation (the same model with no skill) in 47 of 48 held-out judgments, but such rankings cannot show whether any text reads like a person. We therefore built five benchmarks on real writing from before November 2022, rated by blind model judges in piles that mix real and model texts, and tested two uses: writing from notes reconstructed from each real text, and editing it.

Our contributions are:

- An edit benchmark. Asked "Can you improve it?", plain generation changed 76% and 75% of the words of abstracts and appeals and added 3.34 and 2.74 statements per text; version 10.1 of the skill changed 15% and 24% and added 0.12 and 0.29, and 91% and 99% of

ratings judged its edits the writer's own (plain: 0% and 27%).

- A writing-from-notes result: in four genres, held-out model texts stayed separable from real ones (area under the ROC curve, AUC, 0.90 to 1.00; 0.5 is chance), and the skill helped on admitted essays and appeals only.

- Five benchmarks anchored on real human writing, with development and test splits, versions 8 to 10.1 and the released version frozen by content hash before testing, blind single-text judging, and blind checks of unsupported statements and of meaning.

- Copyedit controls showing that fixing real writers' errors barely changed the share judged real, except for English learners, and a detector audit in which at least one of three open-source detectors scored 38 of 40 real admitted essays at 20% or more.

- A released skill (version 13.1: five modes, a genre router, a strict Edit mode), with every earlier version and the evaluation code. On held-out edit texts, judges' 0 to 100 estimate that the writer wrote its "improve it" edits without AI help was 31.7 to 48.2 points above plain generation's and 1.4 to 4.1 below an errors-only copyedit's (detectably only on abstracts); it was untested on writing from notes.

2 Related work

Stylistic signatures of model text. Large language model (LLM) adoption shifted the frequency of style words in abstracts and peer reviews (Kobak et al., 2025; Liang et al., 2024; Juzek and Ward, 2025; Geng and Trotta, 2024). Beyond words, instruction-tuned models write a noun-heavy, informationally dense register (Reinhart et al., 2025), reuse syntactic templates from pre-training data (Shaib et al., 2024), and, in Mandarin, overuse the corrective "not X but Y" construction (Peng et al., 2026). Professional writers agree on a taxonomy of idiosyncrasies in model-written paragraphs, and automated edits built on it improved paragraphs less than expert edits did (Chakrabarty et al., 2025). Syntactic features predict expert "slop" judgments better than lexical ones, though weakly (AUC 0.64 against 0.49) (Shaib et al., 2026). Banning patterns token by token breaks output at about 2,000 patterns (Paech et al., 2026). AI generation leaves a stylometric footprint that AI editing of human text does not reproduce (Shan et al., 2026).

Detecting AI text. Detectors range from trained classifiers (Adam et al., 2026) to zero-shot statistics (Mitchell et al., 2023; Bao et al., 2024; Hans et al., 2024). The RAID benchmark shows that many detectors perform poorly at low false-positive rates and fail under attacks or unseen generators (Dugan et al., 2024), and paraphrasing and simple evasion defeat them (Krishna et al., 2023; Sadasivan et al., 2025; Perkins et al., 2024; Weber-Wulff et al., 2023). Detectors flag unedited non-native English writing as AI at high rates (Liang et al., 2023), one flagged formal human writing at a 76.3% false-positive rate (Creanga and Dinu, 2026), and light AI editing is flagged more often than humanizer output (Karr et al., 2026). Untrained humans detect AI text at about chance (Clark et al., 2021; Jakesch et al., 2023). Watermarks offer a provenance signal that does not depend on style (Kirchenbauer et al., 2023; Dathathri et al., 2024).

AI help with human writing. LLM-written admissions essays differ from those of 30,000 applicants and resemble the essays of privileged applicants (Lee et al., 2025; Alvero et al., 2024); essay style tracked household income before LLMs (Alvero et al., 2021). Estimated LLM use in essays rose sharply in 2024 (Lee et al., 2026), and essays flagged as AI-written were associated with lower admission in one graduate program (Isley et al., 2026). AI writing help homogenizes writing across authors (Padmakumar and He, 2024; Doshi and Hauser, 2024; Agarwal et al., 2025). In grammatical error correction, LLMs tend to make fluency edits beyond the minimal correction (Fang et al., 2023). Our edit benchmark measures how much of a writer's wording and meaning survives a request to improve it.

LLMs as judges. Strong LLM judges agree with human preferences about as often as humans agree with each other, but show position, verbosity and self-enhancement biases (Zheng et al., 2023; Wang et al., 2024), and models can recognize and prefer their own outputs (Panickssery et al., 2024).

3 The skill

The released skill (version 13.1) is a Markdown instruction file of 220 lines (the main file) with six

reference files: a catalog of tells (patterns that readers associate with model-written text) with fixes, guides for personal essays, research, short answers, and messages and appeals, and before/after examples. We built it from two reviews. A survey of 16 open-source writing-skill repositories supplied the procedure of Chen (2026), the "never inject" guardrails of Bronsdon (2026), and the minimum-edit principle and portability test of Yang (2026) (all three under the MIT License). An evidence-graded catalog of tells came from about 40 sources. Appendix A lists every version and its evidence.

Hard rules. The skill never adds a fact, name, number, quote, source, feeling or memory that the user or the source did not provide; where a sentence needs one, it asks or leaves a bracketed question. When editing, it keeps every claim, hedge and number and never makes a claim stronger than its evidence. It uses no evasion tricks (typos, rare synonyms, look-alike characters), makes the minimum edit to human-written text, and coaches application essays instead of writing them.

Audit order. Several well-measured signals are structural (participial tails (Reinhart et al., 2025), corrective "not X but Y" framing (measured in Mandarin by Peng et al., 2026), importance flags, uniform sentence length), and expert readers rely on sentence structure and originality as well as vocabulary (Russell et al., 2025; Shaib et al., 2026). The skill therefore audits a draft in a fixed order: content (the portability test: whether a sentence could move unchanged into someone else's text), staging (announcing a point instead of making it), inflation (ordinary facts presented as significant), stated lessons, rhythm, paragraph endings, tone, vocabulary and format. A strong tell justifies an edit on one sighting; weak tells, such as a single transition word, only when several cluster. A final check looks for tells that rewriting creates, such as uniform rhythm and aphoristic paragraph closers, which a blind test found more often after a humanizer pass (Lenk, 2026).

Modes. *Write* runs the full procedure; for narratives it drafts in two passes, first for shape and then as an editor of its own draft. *Edit* handles the user's own text. *Rebuild*, for drafts the user says an AI wrote, checks every fact, quote and reading against the user's material before writing fresh sentences. *Detect* quotes passages and names patterns without guessing authorship. *Coach*, used for application essays and short answers, interviews the student, proposes a structure, critiques drafts and lists corrections, and does not write the essay or example sentences built from the student's story.

Genre router. A table in the main file (the mode table) sends each request to a mode and one reference file: personal statements and short answers to Coach; research abstracts and reports to Edit of the researcher's own draft; emails, messages and appeals to Write or Edit; other prose to Write or Edit with short genre notes. Each genre guide's main rules came from comparing model writing with real writing of that kind on development data; the messages guide was built from appeals only, and the activity-list advice follows admissions offices' published advice. Since version 12 the mode table tells the agent to read only the matching reference.

Strict Edit mode. The first edit round (Section 5.1) used version 10.1, whose Edit mode asked only for the minimum effective edit that keeps the writer's voice. Version 11, written from the edit benchmark's development halves, makes the minimum edit even when the user says "improve it". It fixes outright errors (spelling, grammar, agreement, articles, punctuation), anything that blocks the meaning and clear AI tells, and keeps the writer's sentences, order, vocabulary, repetitions, paragraphing and punctuation habits. It adds no transitions, thesis preview, definitions, examples, statistics, authorities, idioms or reasons, never makes a claim stronger, and does not make a non-native writer's grammatical phrasing native. After the text it lists up to five bigger changes without writing them; for a learner's practice or graded essay it returns the corrected text and the list, never a rewrite. Versions 12 to 13.1 keep these instructions.

4 Benchmarks

Corpora. The admitted essays (Table 1) were drawn at random from 90 essays that Hamilton, Johns Hopkins, Tufts, MIT and Connecticut College published as admitted applicants' work (Hamilton College Office of Admission, 2026; Johns Hopkins University Office of Undergraduate Admissions, 2026; Tufts University Office of Undergraduate Admissions, 2026; MIT Admissions, 2026; Connecticut College Office of Admission, 2026): 28 personal statements and 12 supplemental essays. The 28 short answers come from 66 answers (40 to 350 words) that MIT and Tufts ap-

Genre	Source	Years	Notes	Edited
Admitted essays	5 colleges	≤ 2022	24/16	–
Short answers	MIT, Tufts	≤ 2022	16/12*	–
Abstracts	ISEF	2014–2021	24/16	16/16
Turkish appeals	charity runs	≤ 2022	32/16	16/15
Learner essays	ELLIPSE	< 2022	–	16/16

Table 1: The five corpora (all English except the appeals): development/test texts written from notes reconstructed from each real text (Notes), and real texts edited (Edited; texts not used for Notes). Years: when the texts were written (essays and short answers: classes of 2026 or earlier). *Model answers were written for invented applicants.

plicants published on admissions blogs (MIT Admissions, 2026; Tufts University Office of Undergraduate Admissions, 2026) and in MIT's student newspaper (The Tech, 2016); we favored international applicants and those in STEM or economics, took at most two per writer, and split them by writer. The abstracts are winning abstracts of the International Science and Engineering Fair (ISEF) from 2014 to 2021 (Society for Science, 2026): 40 drawn at random, stratified by group, from 269 that passed screening (English, 150 to 300 words), and 32 others for editing. The appeals are texts in which runners on a Turkish charity-run donation platform explained, for events up to 2022, why they ran. Cleaning and screening left 84 texts of 60 to 300 words in a first-person voice; the first 32 are development texts, 16 test texts were drawn at random from the rest, and 31 others were used for editing. The learner essays are 32 by English learners in grades 9 to 12 from the ELLIPSE corpus (Crossley et al., 2023) (250 to 500 words, overall proficiency 2.5 to 4.0 of 5, at most three per prompt).

Writing from notes. For each real text, an agent reconstructed the notes the writer could have had beforehand: every fact, number, quote and feeling, regrouped by topic, without the text's order, images or closing lines, and with copying capped at runs of six words for essays, five for appeals and four for abstracts. Each condition (plain generation, skill versions and, for essays, a word-list prompt; Appendix E) wrote the text from the notes alone at the real text's length. Short answers were the exception: an agent refused to turn real admitted students' personal answers into source material for new application answers, so we wrote notes for an invented applicant for each real question (28 applicants from 16 countries). The real answers serve only as the human anchor, so judges may have recognized model answers partly by invented content.

Editing. For the edit benchmark, plain generation and the skill each received every real text with one of two requests, "Can you improve it?" or "Can you fix the grammar and mistakes?", after a one-line framing (a practice essay for English class; a science-fair abstract; a campaign page). Each agent handled two texts as separate requests and returned the revised text only, so the skill's list of suggested changes was not produced. The same model also made an errors-only copyedit of each text, told to fix outright errors and change nothing else.

Judging. Judges read piles of 4 essays, or of 12 to 16 shorter texts, by different writers, never two versions of one writer's text in a pile. Each was told that any mix of real and model texts was possible and gave each text a probability that the writer wrote it (p, 0 to 100; for edits, with no AI writing or editing), a quality score from 1 to 10, and the evidence. Each genre had two judge roles: an editor and an admissions reader (essays), an admissions reader and an essay coach (short answers), an ISEF judge and a journal editor (abstracts), a donor and an NGO editor (appeals), and an English teacher and a writing researcher (learner essays). [generator model, name and version] wrote every text and was every judge except a third essay judge, an admissions reader on [second judge model, name and version], a different model of the same family. Each judge got its own random piles.

Measures. We report the share of ratings with $p \geq 50$ (judged real, or for edits judged the writer's own), mean p, the AUC for telling real texts from a version, and differences paired by text with 95% bootstrap intervals over texts. Absolute shares moved between judging rounds with what else was in the piles: learner-essay copyedits were judged the student's own in 61% of ratings in the first round (both halves) and in 47% and 56% in two later rounds (test halves). Our results are therefore paired differences within a round; we do not compare absolute shares across rounds. Every edit was made by a model, so the share judged the writer's own counts how often judges missed an edit; words changed (share of the original's words changed, averaged over texts; Appendix B) and the meaning check measure how much of the writer's text survived. A blind checker counted unsupported

	ISEF abstracts (32)				Turkish appeals (31)				Learner essays (32)			
Text	Words	Own	p	Add	Words	Own	p	Add	Words	Own	p	Add
Original	–	100%	84.8	–	–	100%	89.9	–	–	100%	95.3	–
Errors-only copyedit	3%	100%	80.6	0.00	6%	98%	83.7	0.00	16%	61%	55.3	0.03
Plain, “fix the grammar”	36%	30%	43.9	0.91	21%	97%	75.0	0.00	38%	3%	30.0	0.81
Skill, “fix the grammar”	10%	91%	69.7	0.09	10%	98%	81.8	0.00	34%	5%	31.9	0.41
Plain, “improve it”	76%	0%	28.6	3.34	75%	27%	40.9	2.74	90%	0%	19.2	3.91
Skill, “improve it”	15%	91%	67.7	0.12	24%	99%	77.1	0.29	47%	3%	30.5	0.62

Table 2: Editing real writing (skill version 10.1, frozen before these texts were collected; development and test halves pooled; two judges per genre, about 64 ratings per text version). Words: share of the original's words changed. Own: share of ratings with $p \geq 50$, where p is the judged probability that the person wrote the text with no AI writing or editing. Add: statements added per text (meaning check). Appendix B gives quality, removed and changed statements, voice, AUC and detector scores.

statements (ones the notes do not support) in each model text. A blind meaning check compared each edit with its original: statements added, removed or changed, and voice kept (1 to 10).

Splits and freezing. Each corpus was split into development and test texts before any judging (edit texts 16/16 per genre; appeals 16/15). Versions were developed on development texts, with two exceptions (see Data reuse in the Limitations). Versions 8, 9 and 10 were frozen, with a content hash recorded, before any test text was written with them, and version 10.1 before the edit and ablation texts, after rewriting the 12 test short answers in its own check. Version 11's changelog records a hash and a freeze before testing, but was written after the test run. Version 13.1 was released, with its hash, before its edit run and was not changed for it. Test texts seen by accident (judge comments on four test essays; notes of five test appeals) were randomly swapped for unseen ones before testing.

5 Results

5.1 Editing real writing

Table 2 gives the first round; every text was new to version 10.1. Asked to improve a writer's text, plain generation rewrote most of it (76% of the words of abstracts, 75% of appeals, 90% of learner essays) and added 2.74 to 3.91 statements per text that the writer never made, such as a thesis preview, a definition, an authority or a stronger conclusion. Paired by text, the skill's “improve” edits scored 39.5 points of p above plain generation's on abstracts [+36.0, +42.9] and 35.9 on appeals [+30.5, +41.7]. Most still changed some meaning: the meaning check found no change in only 6 of 32 abstract edits and 14 of 31 appeal edits (plain generation: none). Judges could still usually tell them from the originals (AUC 0.89 to 0.91 for each judge), and their p was below the errors-only copyedits' (abstracts -12.8 [-17.9, -8.0]; appeals -6.6 [-9.3, -4.3]).

Asked only to fix the grammar, plain generation came close to the skill on appeals (97% and 98% of ratings judged the writer's own; skill minus plain +7.1 [+3.9, +10.7]) but not on abstracts, where its grammar pass rewrote 36% of the words and added 0.91 statements per text (skill minus plain +27.4 [+23.4, +31.5]). On “improve”, the skill's edits were judged 0.28 to 0.94 points lower in quality than plain rewrites (Appendix B); on “fix the grammar”, quality did not differ. English learners' essays were the exception in this round. An errors-only copyedit alone cut the share of ratings judging the essay the student's own from 100% to 61% (Section 5.3). Every model edit changed 34% to 90% of the words, and at most 5% of ratings judged one the student's own (AUC 1.00 for each judge). The skill's “improve” edit still scored 11.3 points above plain generation's [+8.3, +14.2].

Version 11. Version 11 changed only the Edit-mode paragraph, from judges' comments on the development-half learner edits: template transitions (“First”, “Additionally”), definitions and authorities the student had not written, native idioms, and uniformly clean grammar. In a second round, on the test halves, its “improve” edits beat version 10.1's on appeals (+4.1 [+2.3, +6.1]) but not detectably on abstracts or learner essays (Table 7).

The released version against plain generation. In a third judging round of the test halves we ran version 13.1, whose Edit instructions are version 11's, with both requests. Judges rated its edits blind in new piles with the originals, the copyedits and plain edits of the same texts, and we ran

Text	Words	Own	p	Qual.	Add/Rem/Chg
Abstracts (16 texts)					
Copyedit	2%	100%	82.6	6.50	–
Plain	73%	0%	30.2	6.91	3.12/2.50/5.75
v13.1	10%	100%	78.5	6.50	0.00/0.62/0.50
Δp: +48.2 [+44.1, +52.3]; −4.1 [−7.4, −0.9]					
Appeals (15 texts)					
Copyedit	5%	100%	83.7	6.03	–
Plain	69%	30%	42.1	6.53	2.60/1.07/3.87
v13.1	8%	100%	81.4	6.03	0.00/0.00/0.00
Δp: +39.3 [+29.7, +47.9]; −2.3 [−6.3, +1.0]					
Learner essays (16 texts)					
Copyedit	16%	56%	56.6	3.53	–
Plain	93%	3%	23.6	5.19	4.00/3.19/7.94
v13.1	21%	62%	55.2	3.62	0.06/0.00/1.06
Δp: +31.7 [+27.4, +36.0]; −1.4 [−10.0, +7.3]					

Table 3: Released version (13.1), "Can you improve it?", third judging round of the held-out test halves (32 ratings per text version; appeals 30). Columns as in Table 2; Qual.: quality 1–10; Add/Rem/Chg: statements added, removed and changed per text (plain: first-round check of the same edits). Δp: 13.1 minus plain; 13.1 minus copyedit (paired by text, 95% intervals).

Text	Judged	p	AUC	Qual.	Unsup.
Admitted essays, 16 test (48 ratings per version)					
Real	100%	86.4	–	6.21	–
Plain	38%	44.8	0.96	6.38	3.06
Word list	38%	46.7	0.96	6.35	2.69
Skill v7	46%	51.9	0.91	6.29	0.94
Skill v8	65%	59.4	0.90	5.98	0.19
Appeals, 16 test, v9 test run (32 per version)					
Real	100%	91.6	–	4.59	–
Skill v8	12%	37.6	1.00	6.22	0.06
Skill v9	59%	49.3	1.00	6.16	0.00
Same 16, ablation run (32 per version)					
Real	100%	91.2	–	5.22	–
Plain	16%	35.4	1.00	6.25	0.69
Skill v10.1	59%	51.2	1.00	6.16	0.44
ISEF abstracts, 16 test (32 per version)					
Real	100%	87.3	–	5.78	–
Plain	6%	34.5	1.00	6.16	0.00
Skill v10	12%	38.5	1.00	6.06	0.00
Short answers, 12 test (24 per version)*					
Real	100%	88.1	–	4.38	–
Plain	0%	14.2	1.00	7.00	0.58
Skill v10	0%	16.4	1.00	7.04	1.08

Table 4: Writing from notes, held-out test texts. Judged: share of ratings with $p \geq 50$. AUC: real texts against the version (essays: pooled over three judges; other genres: each of two judges, one value when equal at two decimals). Qual.: quality 1–10. Unsup.: unsupported statements per text. Rows from different runs are not directly comparable. *Invented applicants.

the meaning check on them (Table 3). Paired by text, its "improve" edits scored 48.2, 39.3 and 31.7 points of p above plain generation's on abstracts, appeals and learner essays, and 4.1, 2.3 and 1.4 below the copyedits' (only the abstracts interval excludes zero). On "fix the grammar" it scored above plain generation by 28.4 [+22.3, +34.3], 5.9 [+0.8, +12.6] and 15.6 [+5.2, +24.9]. In quality its "improve" edits were below plain rewrites (−0.41 [−0.68, −0.09], −0.50 [−0.77, −0.23], −1.56 [−1.88, −1.22]) and not detectably different from the copyedits' (0.00, 0.00 and +0.09, intervals including zero). The meaning check found no change in 56%, 100% and 31% of them (first-round check of the same texts: plain generation none, learner copyedits 56%). On appeals they changed no statement and were close to the copyedits on every measure. On abstracts and learner essays they stayed near copyedit size (10% and 21% of the words, against 2% and 16%) but changed more statements than the copyedits. Abstract edits removed 0.62 statements per text; the request set a 250-word limit.

For learner essays this qualifies the first round. The released version's edits were not detectably different from the copyedits (−1.4 [−10.0, +7.3]), but in version 11's round, edits made with the same Edit instructions were 11.1 points below them [−21.3, −0.5], so we cannot say whether edits near copyedit size are judged like copyedits. In this round the copyedit alone cost 38.2 points against the untouched essay [−46.0, −30.2].

5.2 Writing from notes

Table 4 gives the held-out results. In every genre, every real text was judged real in every rating, and model texts stayed separable (AUC 0.90 to 1.00) with or without the skill. On essays the two [generator model, name and version] judges separated real essays from version 8's almost perfectly (AUC 0.98); the pooled 0.90 includes the third judge, which called 94% to 100% of every version's essays real. Judges rated model texts higher in quality than the real ones in appeals, abstracts and short answers, and about equal on essays.

The skill helped in two genres. On essays, version 8 (built on 24 development essays) beat plain generation by 14.7 points of p [+9.3, +20.3], the word-list prompt by 12.7 [+6.8, +18.3] and version 7 by 7.5 [+2.4, +13.1]. It had 0.19 unsupported statements per essay against 3.06 for plain generation, with quality −0.31 [−0.58, −0.04] against version 7 and −0.40 [−0.71, −0.04] against plain generation. On appeals, version 9 (built on 32 de-

velopment appeals) beat version 8 by 11.7 points [+5.5, +17.9] with no detectable change in quality, and in a later run on these test appeals, version 10.1 beat plain generation by 15.8 [+8.8, +23.0].

The skill did not help measurably in the other two genres: version 10 scored +4.0 [-0.3, +7.9] over plain generation on abstracts and +2.2 [-1.7, +6.0] on short answers, where no model answer was judged real. Judges rated model short answers above the real ones (7.00 and 7.04 against 4.38) and still caught every one. In the development round they named signs of over-crafting: stacked precise figures and names, comic asides, metaphors set up and paid off, and hedged endings (hedges cited in 50 to 53% of ratings of model answers, none of real ones). Version 10 added more unsupported statements than plain generation (1.08 against 0.58 per answer), mostly by turning the invented applicants' doubts into commitments, which version 10.1 forbids.

5.3 Fixing errors changes judgments little, except for learners

In the first development rounds, judges cited errors in 85% of their ratings of real admitted essays, and errors and non-native phrasing in 56% and 67% of their ratings of real abstracts. Fixing them barely changes the share judged real, though it lowers p. On 24 development essays a copyeditor agent fixed only outright errors (0 to 6 per essay, under 1% of words): every copyedited essay was still judged real, p changed by -1.9 [-3.1, -0.9], and AUC against the originals was 0.61. On 24 development abstracts the copyedits stayed 100% judged real (-0.8 [-2.2, +0.7]; AUC 0.58 and 0.53), and judges still cited non-native but grammatical phrasing in 52% of their ratings. In the first edit round (both halves), copyedits changed p by -3.7 on abstracts [-5.7, -1.8] and -6.4 on appeals [-8.8, -4.3]; 100% and 98% were still judged the writer's own, although judges told copyedited appeals from the originals with AUC 0.81 and 0.77.

Learner essays went the other way: p changed by -40.0 [-45.5, -34.5], and AUC was 0.995 and 1.00. This was also the largest copyedit (16% of words; abstracts 3%, appeals 6%), and not purely errors-only: the meaning check found 0.62 changed statements per essay (none in abstracts or appeals), and no change of meaning in only 20 of 32 essays. To these judges, removing a learner's errors, and possibly some phrasing, removed much of what marked the essay as theirs.

5.4 Which parts of the skill matter

We wrote the test appeals and abstracts from notes with five variants of version 10.1 (Table 8): the full skill, the skill without the tells catalog, the main file alone, the genre reference alone, and a compact main file of 1,094 words. No variant's paired difference from the full skill excluded zero in either genre. On abstracts the full skill did not beat plain generation (+1.4 [-3.2, +5.8]) and was judged lower in quality (-0.38 [-0.69, -0.09]). Unsupported statements differed more (point estimates, 16 appeals, no intervals): 1.06 per appeal without the main file and 0.81 with the compact one, against 0.44 for the full skill and 0.69 for plain generation. This suggests, without a test, that the main file's hard rules and its check of every name, number and claim against the notes carry the no-invention behavior. We detected no effect of the tells catalog (without it minus full skill: abstracts +1.0 [-5.0, +6.6]; appeals +1.1 [-3.7, +6.5]).

5.5 Rankings among model texts

Before the benchmarks on real writing, judges ranked texts of several conditions against each other on invented briefs (Appendix E). On 16 held-out briefs in ten genres, version 4's Write mode ranked above plain generation in 47 of 48 judgments, above a word-list prompt in 43 and above a humanizer skill in 44 (Wilcoxon $p \leq 0.003$ for each, tests over briefs). We chose version 4 over version 6 after seeing these results, judges could see condition folder names, and rankings depended on the panel. On the 16 test admitted essays, version 7, which kept version 4's Write procedure, beat plain generation (+7.1 [+2.7, +12.2]) but not detectably the word-list prompt (+5.2 [-0.2, +10.3]).

Regression on general writing. On the 16 held-out ranking briefs, two judges ranked plain generation, version 9 and version 10.1 blind (32 judgments; Table 9). This design measures preference among model texts, so we use it only to detect damage to general writing. Version 10.1 ranked below version 9 in 28 of 32 judgments and was no better than plain generation (no test over briefs). Version 12 allows contractions in short answers, lets facts be joined into one story while still forbidding invented causes, and tells the agent to read only the matching reference. On the 20 development briefs of the ranking study, with every version given only its genre's reference, version 12 still trailed version 9 (difference in mean rank -0.65

[−1.2, −0.05], positive meaning better; bootstrap over briefs) and was no better than plain generation (+0.12 [−0.7, +0.95]). Version 13 (released as 13.1 after two text corrections) only moves the mode table to the end of the main file. On 12 of these briefs, without plain generation, no difference in mean rank among versions 12, 13 and two independent version 9 samples excluded zero (Appendix D), so this in-sample check cannot show whether version 13 removes the regression.

5.6 Detectors and surface features

We scored texts with three open-source detectors, D1 to D3 (Desklib, 2025; Fakespot, 2025; SuperAnnotate, 2024), described in Appendix E. A text passes at 20 when all three score it below 20% (below which Turnitin shows only an asterisk; Turnitin, 2026) and passes at 50 when all three score it below 50%. On admitted essays the detectors reversed the expected order: on average all three scored real essays above every model version, and at least one scored 38 of 40 real essays at 20% or more. On version 10.1's edits (English only) they agreed with the judges in direction. The skill's "improve" edits of abstracts scored lower than plain generation's on all three detectors (Appendix B), though few passed at 50 (9% against 0%) and none at 20. On learner essays an errors-only copyedit alone moved D1's mean from 11 to 50 and cut the share passing at 50 from 78% to 25%; no model edit passed more than 3%. The detectors also flagged real abstracts: only 34% passed at 50.

A logistic regression on 22 surface features, cross-validated within each genre, separated real from model texts almost perfectly for essays (AUC 0.997; 40 real, 185 model), well for short answers (0.92), and less for appeals (0.80) and abstracts (0.77; 0.64 when trained on the other genres). For abstracts, where judges reached AUC 1.00, these features explain little; the judges' reasons point to content and structure, which we did not measure.

6 Discussion

Edit, don't write. For text a writer must submit as their own, the edit results suggest answering "improve it" with the minimum edit. Its cost is judged quality against plain rewrites (0.41 to 1.56 points for the released version), with no detectable cost against a copyedit. An unchanged text would maximize the judged-own share, so that share must be read with quality and words changed. The released version's edits were near copyedit size, so what the strict mode adds beyond a copyedit rests on its list of bigger changes, which we did not evaluate.

Learners' errors read as voice. Liang et al. (2023) found that detectors flag unedited non-native writing as AI. Our judges did not: every rating judged the uncorrected learner essays the student's own. They did separate the model-made errors-only copyedits from the originals (first round: AUC 0.995 and 1.00), and in the third round the released version's edits did not detectably differ from the copyedits; no edit we tested avoided the cost of correcting a learner's errors. A human tutor's corrections would remove the same errors, so we expect, but did not test, that any fluency-based judgment of AI use would also flag learners given permitted grammar help.

Preference is not human-likeness. Judges rated model short answers above the real ones in quality yet judged none real (Section 5.2); we expect, but did not test, that rules pushing a model toward crafted prose raise its judged quality and make it easier to recognize. Rankings among model texts put version 4 ahead of both baselines, but versions 10.1 to 12 were not detectably ahead of plain generation in later ranking checks, and against real essays only version 7's lead over plain generation was detectable (Section 5.5). Style guidance should be measured against real writing of its genre.

7 Conclusion

Against real pre-2022 writing in five genres, model texts written from notes stayed recognizable to the judges on the generator model, and earlier versions of the skill closed part of the gap only for admitted essays and appeals. Asked to improve a writer's own text, plain generation rewrote it. On held-out texts the released version's edits scored 31.7 to 48.2 points of p above plain generation's and 1.4 to 4.1 below an errors-only copyedit's (detectably only on abstracts). For text the writer must submit as their own, our results suggest that an assistant should fix outright errors, keep the wording and voice, and list bigger changes; we tested this with one generator, model judges and three genres, and did not evaluate the list.

Limitations

No human raters. Apart from a one-person pilot, excluded from analysis, no human rated any text.

The main open question is whether human readers agree with the model judges; a rating survey of texts from the invented briefs is released for this purpose.

Model judges. [generator model, name and version] wrote every text and was every judge except the third judge of the essay benchmark and of the held-out ranking round, which ran on a different model of the same family; models can recognize and favor their own output (Panickssery et al., 2024). Every condition was produced by the same model, so self-preference does not by itself favor one condition, but judges may share the generator's taste, and self-recognition may inflate how well judges separate model texts from real ones. Both judge roles in a genre ran on the same model, so their agreement is not evidence from independent raters. The third essay judge called 94% to 100% of every version's essays real, so separability depends on the judge; in the held-out ranking round, the two same-model judges agreed with each other (mean Kendall τ 0.78) more than with the third judge (0.55 to 0.62). The skill's genre rules (versions 8 to 10) were built from what these judges cited on development texts, so our test sets are held out over texts but not over judges, and a gain may partly reflect these judges' cues rather than what human readers notice. Judge prompts said that any mix of real and model texts was possible, which primes judges to look for AI writing. Judged quality comes from the same model judges and may reward the polish that marks model text, so the quality costs we report for minimum edits may overstate what a human reader would see. Intervals are not corrected for multiple comparisons; we treat single intervals that barely exclude zero as tentative.

Judging runs. Runs drifted, mostly for model texts. On the same 16 test abstracts, plain generation was judged real in 6% of ratings in one run and 22% in another (real abstracts: mean p 87.3 and 86.8; real test appeals 91.6 and 91.2), and the same version 10.1 learner-essay edits were judged the student's own in 3% of ratings in the main edit run (both halves pooled) and 19% of ratings of the test-half edits in the version 11 run, whose piles held no plain rewrites; the learner copyedits also moved across the three rounds that judged the test halves (Section 4). Absolute shares depend on the other texts in a pile, so we compare versions within a run. The exceptions are marked: the two appeal blocks of Table 4, version 10.1's short-answer check, the meaning-check columns of Table 7, and the plain and copyedit meaning-check values compared with Table 3.

Corpora. The real texts were selected by others: colleges chose and often edited the essays they published, ISEF abstracts are winners, and the appeals come from one platform. As public web pages they may be in the models' training data, which could help judges recognize them. The corpora lean toward non-native writers: 14 of the 40 abstracts written from notes and 7 of the 32 edited ones are by Turkish winners, and the short answers were chosen to favor international applicants. The Turkish appeals were judged by models whose reading of Turkish we did not check against native speakers, and the detectors could not score them. The notes behind each model text were reconstructed by a model from the finished text; real notes would be messier, and the notes' spoken register shaped what every condition wrote. Short answers were written for invented applicants, so judges may partly have caught invented content. The copyedits and the meaning check of each edit were made by a model, and the learner copyedits changed some statements (0.62 per essay). Because the real texts stay private, others cannot rerun the benchmarks exactly.

Data reuse. Several versions were developed on the same development texts, so development gains are in-sample. Version 11 was built after we saw pooled (development and test) aggregates of the version 10.1 edit run; its rules came from judge comments on the development halves only. The test appeals and abstracts were reused in the ablation run after earlier tests on them, and the edit test halves were judged in three rounds (versions 10.1, 11 and 13.1); the released version's Edit instructions were not changed after version 11's test (version 13.1 corrected one evidence figure in that section; no instruction changed). Version 10.1's short-answer check followed a reading of version 10's test report. The 16 held-out ranking briefs, also used to choose version 4 over version 6, became development data after the regression. The 20 development briefs of the ranking study were reused for the version 12 and 13 checks.

Scope. One generator model wrote every text. Results against real writing were measured with versions 7 to 11 and, on the edit benchmark's held-out halves only, with the released version 13.1. The release was not tested on writing from notes:

apart from the edit run, only its predecessor version 13 (same instructions) was checked, on 12 development ranking briefs, without plain generation, where it could be told apart from neither version 9 nor version 12, and in a five-request routing probe that version 12 also passed. Its short-answer changes were not tested against real short answers, and the detectors were not run on its edits. Neither the word-list prompt nor the humanizer was run on the edit benchmark, and the humanizer was never judged against real writing. The skill's list of suggested changes was not produced in the edit benchmark and was never rated, and Edit mode was not ablated. No plain condition was asked for the smallest possible edit, so "fix the grammar" is the closest minimal-instruction baseline. The ranking study used fictional, model-written student personas, and Coach mode was tested only with simulated students and version 6's personal-essay guide (Appendix F), which has changed since; short-answer coaching was not tested. Each condition produced one text per item, so a result on an item also reflects the luck of that sample. In the first edit round, a safety filter repeatedly refused one judge pile of abstract edits, so ratings per abstract version there are uneven (60 to 67; 64 planned), and in the version 11 development run abstracts had 20 to 30 of 32 ratings per version. We do not report the share of words changed for version 11. Commercial detectors used by schools were not tested, and the detector results rest on small, selected human samples.

Ethical considerations

Dual use. Earlier versions of the skill often lowered detector scores (version 4 on invented briefs, version 10.1 on edited abstracts; Section 5.6; Appendix E), and a student could use the skill to pass off model-written work as their own. Minimum edits of abstracts and appeals were also rarely identified as AI-edited: in each edit round at least 91% of ratings judged the skill's "improve" edits the writer's own, and every rating of the released version's did. The detectors were not run on the released version's edits. The released version uses no evasion tricks (typos, homoglyphs, invisible characters) of the kind studied in the attack literature (Krishna et al., 2023; Sadasivan et al., 2025), but on six held-out argument, explainer and literary-analysis briefs every plain text was flagged by at least one detector and every text version 4 wrote passed all three. Its mode table sends school essays to Coach when they will be graded as the student's own work, which it learns only from the user; our evaluation prompts asked for Write mode or said the students were fictional, and got finished essays. For application essays it coaches and does not write, and it tells the student that the Common App, Yale and many programs count substantive AI-written content as not the applicant's own work (Common App, 2023; Yale Undergraduate Admissions, 2026). It tells users that recent models of the generator's family add a provenance watermark (Anthropic, 2026; BleepingComputer, 2026) and that the skill is not a way to hide AI use.

Detector developers. Because we report low detector scores for model text and high scores for human essays, we [notified the developers of all three detectors on <date>], at least 30 days before any preprint.

Real writers' texts. The five corpora contain real people's writing, much of it by school students. We release only counts and aggregate scores; the real texts and the model texts derived from them stay private. The released detector score files identify essays by source and number, not by writer, so no named person is linked to an AI score. We never quote a text from these corpora in this paper. When an agent refused to turn real students' personal answers into material for new application answers, we accepted the objection and redesigned the short-answer benchmark around invented applicants (Section 4).

Fairness to learners. Our judges and detectors rated learner essays as more likely AI-written or AI-edited once a model had corrected their errors (Sections 5.3 and 5.6), so neither should be read as a test of authorship, least of all for non-native writers.

Human raters. A planned human rating study, not reported here, uses volunteers who give informed consent, collects no names or contact details, and [has approval from <school or teacher>; keep it unnamed in the review version]. Raters under 18 take part only with a parent's or guardian's permission, which the survey asks for but cannot verify. The generated texts from our invented briefs and the survey's answer key are public, so a rater who searched for a passage could learn its condition; the survey asks raters not to.

AI use statement

AI agents ([generator model, name and version] and [second judge model, name and version]) did a large share of this work: they surveyed the literature and existing skills, drafted every version of the skill, generated all evaluation texts, acted as judges, wrote the analysis scripts, and drafted this paper. The development log, prompts, scripts and every intermediate version are released. [The authors describe here, accurately, what they did themselves: e.g. which decisions they made, which numbers and citations they checked against the data and sources, and which sections they rewrote.]

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A Version history

Table 5 lists every version. Versions 1 to 4 were shaped on a seven-brief pilot; versions 5 and 6 on 20 development briefs; versions 8 to 11 against real writing; version 12 on the 16 held-out ranking briefs after the regression check, and version 13 on the 20 development briefs. Version 13.1 corrected two lines of text and was then run, unchanged, on the edit benchmark's test halves (Section 5.1). Before the genre guides of versions 9 and 10, the general skill did not help on development texts in either new genre (appeals: version 8 -2.2 [-5.2, +0.8] against plain; abstracts: version 9 -3.6 [-8.0, +0.3]), partly because its own rules added tells: semicolons in appeals (1.22 per text against 0.41 in real appeals), and glossed terms and conversational simplification in abstracts. Version 13 was also checked with a routing probe: five invented requests, two samples each, graded blind, all routed to the intended mode. Version 12 also passed 10 of 10. The probe ran with a project instruction file that itself says to coach application essays, so its two Coach requests test the skill less than the other three.

Version	Main change	Developed on	Evidence (set; development or held-out)
v1	audit while drafting	7-brief pilot	personal statements vs plain 0:4 (development)
v2	stricter sample-essay rules	pilot	0:4 (development)
v3	countable checks	pilot	0:4; unsupported 1 vs 16 (development)
v4	two-pass drafting	pilot	1:3; unsupported 6 vs 19 (development); held-out briefs: mean rank 1.92 of 6, above plain in 47 of 48
v5	plain student voice, Rebuild mode, personal-essay checklist	20 development briefs	below v4 in 28 of 40 judgments ($p = 0.055$, development); not released
v6	rules from a diagnosis of development judgments	20 development briefs	above v4 in 21 of 48 judgments ($p = 0.35$, held-out briefs); not released
v7	v4 Write plus v6 Rebuild, example hygiene	assembled from v4 and v6	smoke test on 3 held-out drafts
v8	written (not spoken) voice, one whole-hearted conclusion, check for engineered structure, no added connective observations	24 development essays	essay test: +7.5 [+2.4, +13.1] over v7; unsupported 0.19 vs 0.94
v8c	trying-hard voice, scenes, attempted images	24 development essays	development: -1.7 [-5.1, +1.7] vs v8; unsupported 0.58 vs 0.12; rejected
v9	guide for messages and appeals, punctuation by register	32 development appeals	appeal test: +11.7 [+5.5, +17.9] over v8
v10	genre router, research and short-answer guides	24 development abstracts, 16 development short answers	test vs plain: abstracts +4.0 [-0.3, +7.9], short answers +2.2 [-1.7, +6.0]
v10.1	conclusions no stronger than the writer holds, no invented causal links	v10's short-answer test report	0.25 vs 1.08 unsupported per short answer (test items rewritten, not re-judged)
v11	strict Edit mode	16 development texts per edit genre	edit test vs v10.1 ("improve"): appeals +4.1 [+2.3, +6.1], abstracts +0.2 [-3.1, +3.6], learner essays +1.8 [-6.1, +10.7]
v12	read only the matching reference; facts may be joined into one story; contractions allowed in short answers	16 held-out ranking briefs (regression)	20 development briefs: above v9 in 13 of 40 judgments; vs v9 -0.65 [-1.2, -0.05]; vs v11 +0.23 [-0.35, +0.8]; vs plain +0.12 [-0.7, +0.95]
v13	mode table moved from the top of the main file to the end	20 development briefs (v12 diagnosis)	12 development briefs vs two v9 samples: -0.25 [-1.33, +0.75], -0.21 [-1.00, +0.58]; vs v12 +0.46 [-0.38, +1.25]; v12 on the same briefs vs v9: -0.71 [-1.71, +0.25], -0.67 [-1.54, +0.29] (in-sample); routing probe 10 of 10 (v12 also 10 of 10)
v13.1	two text corrections in the main file (an evidence range; a pointer to the genre notes); no instruction changed	–	edit test halves, "improve it", vs plain: abstracts +48.2 [+44.1, +52.3], appeals +39.3 [+29.7, +47.9], learner essays +31.7 [+27.4, +36.0]; vs errors-only copyedit -4.1 [-7.4, -0.9], -2.3 [-6.3, +1.0], -1.4 [-10.0, +7.3]; released

Table 5: Every version of the skill. Differences in p are paired by text with 95% bootstrap intervals; rank differences for v12 and v13 are differences in mean rank, bootstrapped over briefs; positive means the row's version ranked better. v1–v4 rows: personal-statement pairwise judgments of better writing in development (skill wins : plain wins; four judgments per row), with unsupported claims summed over the same pairs. v3–v4 used depth notes (the student's answers to the skill's depth questions); v4 on the original notes went 1:1 with 4 vs 14 unsupported claims; v1–v2 used the original notes, where unsupported claims were 16 vs 29 and 6 vs 33. Content hashes (Section 4 says which were recorded before testing): v8 8fe68e27a7a0e007, v9 6e56b690fba3cd03, v10 d05f855dc7111b57, v10.1 3571b11768657399, v11 cdda4484b09a1155, v12 0fd7410cda39fa06, v13 5e5db96e61ff8552, v13.1 5f1eae3111a96823.

B Edit benchmark: full results

Table 6 completes Table 2, and Table 7 gives the version 11 comparison. Words changed comes from a word-level alignment (Python's difflib, junk heuristic off) that counts the longer of the original and edited span for every changed span; the total is divided by the original's word count. Words are whitespace-separated tokens, so a punctuation fix changes the word it is attached to. Plain generation's "fix the grammar" pass changed 2.91 statements per abstract, against 0.47 for the skill. Judges separated plain generation's "improve" rewrites from the originals perfectly in every genre (AUC 1.00 for each judge). On "improve", version 10.1's edits were judged lower in quality than plain rewrites (abstracts -0.28 [-0.47, -0.09], appeals -0.41 [-0.61, -0.23], learner essays -0.94 [-1.17, -0.70]) but above the originals on abstracts and appeals (6.42 against 6.11; 6.10 against 5.57), and against the copyedits they gained 0.25 [+0.08, +0.42] and 0.35 [+0.16, +0.58] quality points on abstracts and appeals. On "fix the grammar", quality did not differ from plain generation's (abstracts 0.00 [-0.22, +0.22]; appeals -0.07 [-0.26, +0.14]).

In the version 11 run, whose piles held no plain rewrites, version 11's test "improve" edits were judged the writer's own in 100% of ratings on abstracts and appeals and 28% on learner essays (version 10.1: 100%, 100% and 19%; copyedit: 100%, 97% and 47%). Against the copyedits their p differed by -5.0 [-7.7, -2.4], -1.2 [-3.2, +1.1] and -11.1 [-21.3, -0.5]. On appeals they were not judged better than the originals in quality (+0.13 [-0.20, +0.43]), and the meaning check found no change in any of them (voice kept 10.0), so they were close to errors-only copyedits. On "fix the grammar", version 11 gained on test abstracts (+2.5 [+0.4, +4.8]), a single interval that barely excludes zero.

In the third round (Table 3), the copyedits differed from the originals by -3.5 [-5.7, -1.4] on abstracts, -6.0 [-10.1, -2.9] on appeals and -38.2 [-46.0, -30.2] on learner essays. Judges told the released version's "improve" edits from the originals with AUC 0.82 and 0.71 on abstracts, 0.79 and 0.79 on appeals, and 0.99 and 1.00 on learner essays (one value per judge). Its "fix the grammar" edits differed from the copyedits by -8.0 [-12.6, -4.0], -2.7 [-5.7, +0.1] and -2.4 [-11.3, +7.2]. The meaning check of these edits found 0.06 changed statements per abstract and none added or removed, no change in any appeal, and 0.19 added and 1.00 changed per learner essay.

Genre	Text	Quality	Stmts. rem. / chg.	Voice	AUC (two judges)	D1 / D2 / D3	Pass
ISEF abstracts	Original	6.11	–	–	–	18 / 28 / 44	34%
	Errors-only copyedit	6.17	0.00 / 0.00	10.0	0.64, 0.64	22 / 33 / 50	28%
	Plain, “fix the grammar”	6.34	0.25 / 2.91	7.4	1.00, 1.00	63 / 62 / 70	3%
	Skill, “fix the grammar”	6.36	0.00 / 0.47	9.4	0.82, 0.87	35 / 45 / 65	6%
	Plain, “improve it”	6.67	2.31 / 5.91	5.1	1.00, 1.00	63 / 66 / 70	0%
	Skill, “improve it”	6.42	0.66 / 1.19	8.9	0.91, 0.89	37 / 42 / 62	9%
Turkish appeals	Original	5.57	–	–	–	–	–
	Errors-only copyedit	5.78	0.00 / 0.00	10.0	0.81, 0.77	–	–
	Plain, “fix the grammar”	5.94	0.00 / 0.55	8.9	0.92, 0.87	–	–
	Skill, “fix the grammar”	5.92	0.00 / 0.06	9.8	0.83, 0.82	–	–
	Plain, “improve it”	6.55	1.03 / 3.52	6.0	1.00, 1.00	–	–
	Skill, “improve it”	6.10	0.26 / 0.68	8.7	0.89, 0.91	–	–
Learner essays	Original	2.97	–	–	–	11 / 2 / 22	78%
	Errors-only copyedit	3.75	0.00 / 0.62	9.4	0.995, 1.00	50 / 16 / 45	25%
	Plain, “fix the grammar”	4.42	0.62 / 4.38	6.9	1.00, 1.00	84 / 76 / 71	0%
	Skill, “fix the grammar”	4.23	0.28 / 2.91	7.9	1.00, 1.00	77 / 68 / 66	3%
	Plain, “improve it”	5.44	2.72 / 8.31	4.3	1.00, 1.00	98 / 94 / 72	0%
	Skill, “improve it”	4.50	1.00 / 3.69	7.0	1.00, 1.00	84 / 74 / 70	0%

Table 6: Edit benchmark, skill version 10.1, continued from Table 2. Quality: 1–10. Stmts. rem. / chg.: statements removed and changed per text (blind meaning check). Voice: writer’s voice kept, 1–10. AUC: each text version against the originals, for each judge (abstracts: journal editor, ISEF judge; appeals: donor, NGO editor; learner essays: writing researcher, teacher). D1–D3: mean detector scores (0–100, probability of AI); Pass: share passing at 50 (all three below 50). The detectors cannot score Turkish.

Genre	Split	Δp, improve	Δp, fix grammar	Δ qual., improve	No meaning change	Voice kept
Abstracts	dev	+7.3 [+3.1, +12.6]	+2.3 [−1.8, +7.2]	−0.03 [−0.23, +0.13]	31% / 56%	9.1 / 9.1
	test	+0.2 [−3.1, +3.6]	+2.5 [+0.4, +4.8]	−0.12 [−0.41, +0.16]	6% / 44%	8.7 / 9.1
Appeals	dev	+4.0 [+1.6, +6.3]	+1.1 [−0.3, +2.6]	−0.31 [−0.66, −0.03]	50% / 94%	8.7 / 9.8
	test	+4.1 [+2.3, +6.1]	+1.7 [−1.0, +4.3]	−0.30 [−0.47, −0.13]	40% / 100%	8.7 / 10.0
Learner essays	dev	+8.4 [+2.7, +14.0]	+3.5 [−0.9, +8.0]	−0.41 [−0.66, −0.19]	0% / 31%	7.2 / 8.4
	test	+1.8 [−6.1, +10.7]	+4.4 [−1.3, +10.4]	−0.56 [−0.81, −0.31]	0% / 44%	6.9 / 8.9

Table 7: Version 11 against version 10.1 on the edit benchmark, development and test halves. Δ: version 11 minus version 10.1, paired by text, with 95% intervals; paired over 16 texts, except abstracts development and appeals test (15). No meaning change: share of “improve it” edits in which the meaning check found no change (version 10.1 / version 11). Voice kept: on “improve it”, 1–10 (version 10.1 / version 11). In these two columns, version 10.1 values come from the first meaning check and version 11 values from a separate check run of the same design.

C Admitted essays: developing version 8

A first design, finding the real essay among four essays telling one story, failed because the three model essays shared paraphrases of the same notes (all 20 judgments found it), so we switched to single-text piles. Every text was normalized (straight quotes, single spaces, no title). On the 24 development essays, version 7’s essays differed from the real ones in ways its own rules amplified. They used “really” 21.0 times per 10,000 words, against 4.7 in real essays and 13.7 in plain generation, and “kind of” 12.1 times (real essays never). They had twice the contractions and shorter sentences (15.5 against 17.9 words), and stated a lesson 0.2 times per essay against 1.2. Told to sound like “a smart teenager talking” and not to write “taught me”, the model carried the notes’ spoken register into the essay and ended on a shrug (“I don’t actually know.”) where admitted students ended on a conviction. In the first round (360 ratings), every real essay was judged real in every rating, the two [generator model, name and version] judges put every real essay above every model essay (AUC 1.00), and the real essays got the lowest quality scores (5.97 against 6.14 to 6.42).

Coder agents, blind to version, marked which of 16 cues each judge rationale cited and in which direction. Tidy structure (a neat arc, motifs and callbacks, everything tied together) was cited as a sign of a model in 75% to 86% of ratings of every model version and in 6% of ratings of real essays. A stated moral was cited as a sign of a real student in 54% of ratings of real essays and as a sign of a model in about half of ratings of model essays.

The development rounds produced three candidates. The first (write, don't talk; state the conclusion) removed the filler but turned stiff, and its round-one gain did not replicate. The second, released as version 8, kept those changes, loosened the structure, stopped adding links between scenes that the notes do not make, and required one wholehearted conclusion; in round two it beat version 7 by 6.3 points of p [+2.2, +10.5] and cut the share of ratings citing tidy structure from 86% to 51%. The third (8c) added a trying-hard voice, scenes and attempted images: mean p tied with version 8 (-1.7 [-5.1, +1.7]) and quality rose 0.18, but it invented about five times as many small details (0.58 against 0.12 unsupported statements per essay; version 7 had 1.50), so we froze version 8 before any test essay was written with it. On the 16 test essays, the two [generator model, name and version] judges rated 47% of version 8's essays real, against 22% of version 7's and 9% of plain and word-list essays; their mean p for version 8 was 11.4 points above version 7's [+4.5, +18.7]. On the 16 test essays, version 8 also passed all three detectors less often than version 7 (6 of 16 against 10), plausibly because it moved toward the register of the real essays, which the detectors flag most.

D Ablations and regression checks

Table 8 gives the ablations and Table 9 the regression checks. In the first regression check every reference file was supplied with every brief. In the version 12 check every version, version 9 included, was given only its genre's reference, so no genre's rules could carry over to another, yet versions 11 and 12 still trailed version 9; that check therefore tested the loosened short-answer rules and did not test the instruction to read only the matching reference. In it, plain generation ranked first in every judgment of its eight personal-statement and short-answer briefs. On the 12 development briefs (arguments, explainers, literary analyses), version 12 ranked below each version 9 sample in 16 of 24 judgments, and version 13 in 14 and 13. The compact main file was less than half as long as the full one (1,094 against 2,634 words) but had more unsupported statements on appeals, so we kept the full skill.

	Abstracts		Appeals	
Variant	Real / p	Uns.	Real / p	Uns.
Plain	22% / 39.2	0.19	16% / 35.4	0.69
Full skill	22% / 40.7	0.06	59% / 51.2	0.44
No tells catalog	38% / 41.7	0.00	56% / 52.3	0.50
Main file only	44% / 45.2	0.06	59% / 53.0	0.50
Genre reference only	28% / 42.8	0.12	50% / 50.9	1.06
Compact main file	25% / 39.8	0.06	44% / 48.8	0.81

Table 8: Ablations of version 10.1, writing from notes on the 16 test items per genre (32 ratings per variant), in a judging run separate from Table 4 (there, plain abstracts were judged real in 6% of ratings, mean p 34.5). Real: share judged real; Uns.: unsupported statements per text. Every variant's paired difference from the full skill has an interval that includes zero.

Version	Rank	First	Qual.	Human	Uns.
16 held-out ranking briefs, 32 judgments					
Plain	2.22	11	6.91	5.19	1.31
v9	1.47	18	7.59	6.31	1.12
v10.1	2.31	3	6.81	5.72	1.25
20 development briefs, 40 judgments					
Plain	2.70	16	7.08	5.25	1.10
v9	1.93	15	7.62	6.00	1.73
v11	2.80	3	6.92	5.50	1.90
v12	2.58	6	7.00	5.62	1.27
12 development briefs, 24 judgments					
v9, sample 1	2.25	10	7.71	5.25	3.08
v9, sample 2	2.29	6	7.67	5.29	3.17
v12	2.96	4	7.12	5.12	2.96
v13	2.50	4	7.54	5.17	2.92

Table 9: Regression checks in the ranking design (Appendix E): mean rank among the texts of a brief (1 = best), first places, quality and human-likeness rating (1–10), unsupported claims per text. Top: the 16 ten-genre briefs that were held out in the ranking study. Middle and bottom: its development briefs (bottom: arguments, explainers and literary analyses, with two independent version 9 samples). Bottom, differences in mean rank (positive: first version better; bootstrap over briefs): v13 vs the v9 samples -0.25 [-1.33, +0.75] and -0.21 [-1.00, +0.58]; v12 vs the v9 samples -0.71 [-1.71, +0.25] and -0.67 [-1.54, +0.29]; v13 vs v12 +0.46 [-0.38, +1.25]; v9 sample 2 vs sample 1 -0.04 [-0.92, +0.79].

E Earlier model-versus-model evaluation

We evaluated the skill in three rounds, after a seven-brief pilot that shaped versions 1 to 4: a development round on 20 briefs, a version comparison on the same briefs, and a held-out round on 16 new briefs written after versions 4 and 6 were frozen.

Briefs. The development set has 20 briefs in five genres, four each (Table 11). Four fictional students, each with about 1,100 to 1,600 words of interview notes and answers to the skill's depth questions, supply the material for a Common App personal statement and a 250-word supplemental answer. The other briefs are 11th-grade argument essays and science-club explainers with no facts file, and literary analyses restricted to four passages from a public-domain novel or play, copied from Project Gutenberg and checked by script. The held-out set has 16 briefs: two new students (a personal statement and a short answer each), two arguments, two explainers, two literary analyses, and six briefs in five genres outside the development set. Every brief names the only permitted source of specifics; the people in the facts files are fictional.

Conditions. Fresh [generator model, name and version] agents wrote every text. Plain (P): the model's default writing. Word list (L): the same request plus an instruction to avoid 24 common AI words, six stock phrases and constructions, and em dashes and to vary sentence length. Humanizer edit (H): the most-starred of the 11 humanizer skills we surveyed (about 59,000 GitHub stars on 20 October 2026) (Chen, 2026), applied to P's draft. The skill conditions were the Write mode, the Edit mode applied to P's draft, and from version 5 on a Rebuild mode. On briefs about fictional people, the skill and humanizer prompts said the people were fictional, so that the skill would write the piece instead of switching to Coach mode.

Judges. Judge agents read all texts for a brief in a shuffled order under neutral labels, but each judge opened the texts by file path, and the path contained the condition's folder name. Blinding was therefore imperfect: in 6 of the 128 writing-panel judgments the explanation names a text's condition letter. The judges were a strict editor and the genre's real reader. Each ranked the texts, scored each from 1 to 10 on quality, human-likeness, specificity, voice and structure, rated AI suspicion from 1 to 5, and quoted every claim not supported by the facts file (or every invented statistic, source or factual error when there was none), a coarser, judge-based version of checking generated facts against a knowledge source (Min et al., 2023). In the held-out round a third judge ran on [second judge model, name and version]. We report mean rank with bootstrap intervals over judgments, a Friedman test on per-brief mean ranks, and pairwise Wilcoxon signed-rank tests on per-brief mean ranks, so the unit of every test is the brief (16 held-out), not the judgment (48). Resampling briefs instead of judgments widens version 4's held-out interval from [1.62, 2.23] to [1.52, 2.38]. Over the 15 held-out pairwise tests, Holm's correction keeps every comparison we report as significant below an adjusted p of 0.025.

Held-out results. Table 10 and Figure 1 give the result. We chose which of versions 4 and 6 to release after seeing these results, so version 4's figures are the better of two candidates. Version 4's Write mode ranked first in 23 of 48 judgments and had a mean rank of 1.92 out of 6, against 5.65 for plain generation (Friedman $\chi^2 = 51.0$, $p < 10^{-8}$). The Rebuild mode, applied to the plain draft, ranked above the humanizer applied to the

Condition	Rank	1st	Human	Unsup.
Plain	5.65	0	4.48	3.38
Word list	4.17	3	6.48	2.98
Humanizer edit	4.25	2	5.69	1.88
Skill Write (v4)	**1.92**	**23**	**7.15**	1.17
Skill Write (v6)	2.42	17	6.96	0.96
Skill Rebuild (v6)	2.60	3	7.10	**0.79**

Table 10: Held-out ranking round: 16 briefs, three judges each (48 judgments). Rank: mean rank among six texts (1 = best); 1st: first places; Human: human-likeness rating (1–10); Unsup.: claims per text that judges found unsupported by the facts file or wrong.

same draft in 42 of 48 judgments ($p = 0.0008$) and had the fewest unsupported claims (0.79). Version 6, which added rules derived from a diagnosis of the development judgments, ranked above version 4 in 21 of 48 judgments ($p = 0.35$); one judge called its Rebuild of an argument brief "arithmetic-heavy for an English class." The third judge, on a different model, gave the same order (version 4 1.94, plain 5.38). By genre, version 4 ranked best on personal statements (1.33), arguments (1.33), literary analysis (1.33), the emails (1.67) and the news article (1.00); the word-list prompt beat it on the blog post (1.67 against 4.67) and narrowly on the cover letter (2.00 against 2.33). With one or two briefs per genre these are descriptive.

Development round and version comparison. On the 20 development briefs, version 4's Write mode ranked first in 28 of 40 judgments (mean rank 1.45 of 5; plain 4.62; word list 2.27; humanizer 3.48; Edit mode 3.17). Version 5 (a plain student voice for school essays, a Rebuild mode, a personal-essay checklist) raised the human-likeness rating (1–10; 7.45 against 6.90) but ranked below version 4 in 28 of 40 judgments ($p = 0.055$): judges caught simplified reasoning, and the rule's example stance line became the opening of three of four argument essays. The same plain and version 4 personal statements, judged in two development panels, went 8 to 0 for version 4 in one and 3 to 5 in the other; the first panel also held two edits of the plain draft, and judges may have marked the whole family down. The held-out panels again include an edit of the plain draft (44% shared word 6-grams), so version 4's held-out lead on personal statements and short answers (12 of 12 judgments) may be inflated the same way.

Detectors on invented briefs. D1 is a DeBERTa-v3-large model trained on RAID (Desklib, 2025), D2 a RoBERTa-base model (Fakespot, 2025), and D3 a RoBERTa-large model (SuperAnnotate, 2024). All 16 held-out texts written with version 4's Write mode and all 16 version 6 rebuilds scored below 20% on all three detectors, against 9 of 16 for plain generation and 12 each for the word-list prompt and the humanizer edit. The development round agreed (17 of 20 Write-mode texts against 9 of 20 plain; the Edit mode passed 10 of 20). On personal statements and short answers every condition passed, with no score above 3% on the held-out briefs. D3 was trained on a RAID subset plus generations that include output from the generator's model family (SuperAnnotate, 2024). Texts are split into chunks of about 350 words, and document scores are length-weighted means. As human baselines we scored 16 admitted essays (Hamilton College Office of Admission, 2026; Connecticut College Office of Admission, 2026), 18 high-scoring PERSUADE 2.0 essays (Crossley et al., 2024) and three excerpts from two science explainers published in 2020: none of the 16 essays scored below 20% on D3 (mean 61%), and 2 of the 18 PERSUADE essays passed all three (Figure 2).

Personal statements: arc against fidelity. Personal statements were the hardest genre in this study (Table 5). Plain generation built a better arc by writing connective feelings and causes the student never stated (3 to 11 unsupported claims per essay), and judges preferred it. Forbidding invention flattened the skill's first three versions; one judge wrote that a version 3 essay "mostly walks through the notes in order." Two-pass drafting in version 4 recovered part of the arc, with 4 unsupported claims against 14 on the original notes. On the two held-out students, version 4 ranked first in 5 of 6 personal-statement judgments with 0.67 unsupported claims per essay, against 4.83 for plain generation.

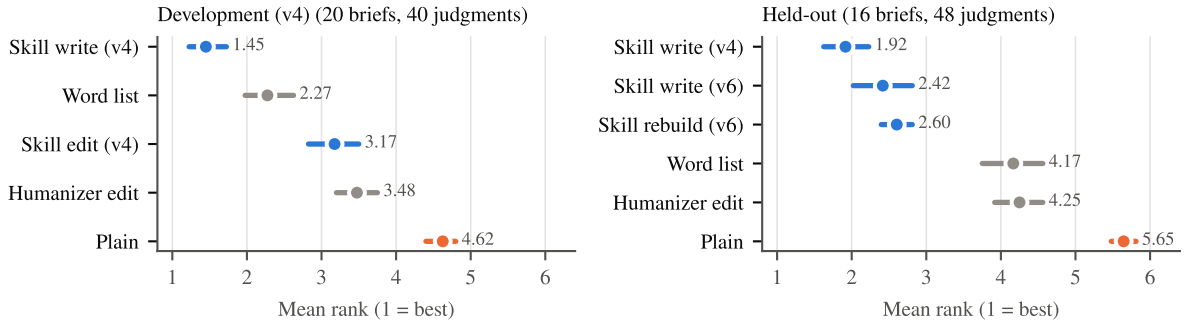


Figure 1: Mean rank with 95% bootstrap intervals (1 = best). Left: development briefs, five texts per panel. Right: held-out briefs, six texts per panel, three judges each. Blue: the skill; orange: plain generation; grey: other approaches.

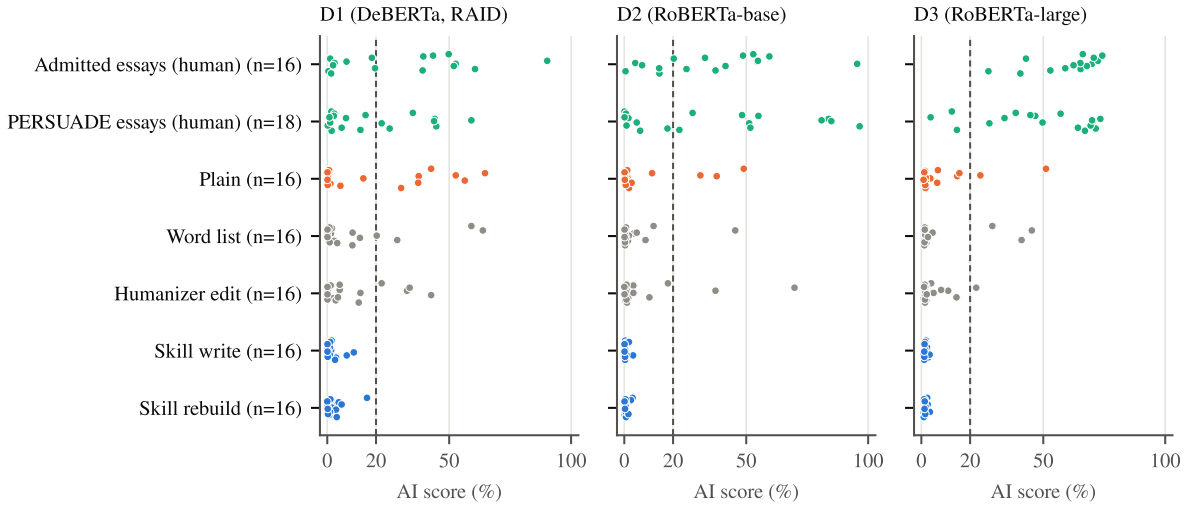


Figure 2: Detector scores per text on the held-out briefs of the ranking study (Skill write: version 4; Skill rebuild: version 6), with human writing collected before November 2022 for comparison (green). D1–D3 are the detectors of Section 5.6. The dashed line is 20%. D3's scores saturate near 73%.

Set	Genre (n)	Topics
Development	Personal statement (4)	Four fictional students (Portuguese-American shoe-repair family; Nebraska church organist; Kazakh wild-apple collector; a Houston lion-dance performer), Common App prompts
	Short answer (4)	Same students, 250-word supplements (curiosity, roommate note, activity)
	Argument (4)	Phone bans; test retakes; group-project grades; part-time jobs
	Explainer (4)	mRNA vaccines; seasons; photosynthesis; sky color and sunsets
	Literary analysis (4)	The Great Gatsby; Frankenstein; Macbeth; The Scarlet Letter (four verified quotes each)
Held-out	Personal statement (2)	Two new students: a Polish e-scooter repairer; an Arizona tamale-stand and weather-station student
	Short answer (2)	Why engineering; the world you come from
	Argument (2)	Later school start; graded homework
	Explainer (2)	Noise-cancelling headphones; leap years
	Literary analysis (2)	Romeo and Juliet; Pride and Prejudice
	Email (2)	Extension request to a teacher; cold email to a professor
	Cover letter (1)	Junior counselor at a library coding camp
	Blog post (1)	Robotics team at a regional competition
	Lab discussion (1)	Pendulum period against length
	News article (1)	Reusable cafeteria trays

Table 11: Writing briefs of the ranking study. Every brief names the only permitted source of specifics (interview notes, a facts file, four quotes, or none for arguments and explainers). All people are fictional. The held-out briefs were written after version 6 was fixed; they became development data after the regression check (Section 5.5).

F Coaching experiment: details

For each of the six fictional students, an agent playing the student wrote a first draft of the personal statement from their notes, with no style guidance. The draft then received feedback from the skill's Coach mode (version 6 files) or from the plain model asked for feedback, and the student agent revised it once with each. Two blind judges ranked the first draft and both revisions, and a counselor agent rated both pieces of feedback. Both kinds of feedback improved the first drafts: the draft ranked last in all 12 judgments. The revision made with Coach-mode feedback ranked above the one made with plain feedback in 7 of 12 judgments (not significant), with fewer unsupported claims (1.67 against 2.75) and lower AI suspicion (1.83 against 2.08). The counselor judge preferred Coach mode's feedback for all six students (usefulness 8.67 against 7.17 of 10) and found less invented content in it (1.17 items per student against 5.17). One plain response supplied what the counselor called "the essay's interpretive center," a line the student never said, and framed it as his own answer; the counselor counted this as substantive content of the kind admissions policies forbid (Common App, 2023; Yale Undergraduate Admissions, 2026). The simulated students copied almost nothing from either feedback (on average under 1% of word 6-grams; at most 3.7% in one revision). Coach mode's responses were about 1.5 times longer (2,264 words against 1,468), which may have helped them with the counselor, since model judges tend to favor longer answers (Zheng et al., 2023). The feedback comparison rests on one counselor judgment per student, from a model that could see each condition's folder name in the file path, and four of the six students are development personas.